National
Qualifications
2025

X819/76/11 Design and Manufacture

TUESDAY, 13 MAY
1:00 PM – 3:15 PM

Total marks — 80

SECTION 1 — 25 marks

Attempt ALL questions.

SECTION 2 — 55 marks

Attempt ALL questions.

Write your answers clearly in the answer booklet provided. In the answer booklet you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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X8197611

SECTION 1 — 25 marks

Attempt ALL questions

1. Two equipment trolleys are shown below with product information.



Medical trolley

Materials

stainless steel shelves

tubular stainless steel frame

nylon castor wheels

Additional details

360° swivel wheels

Tool trolley

Materials

ABS shelves and handles

mild steel side panel

polypropylene tool hooks

galvanised mild steel wheel brackets

Additional details

360° swivel wheels

self-assembled



MARKS

1. (continued)

(a) Explain why the materials chosen are suitable for the trolleys. **6**

(*You must give **six** different explanations*.)

(b) Name **three** appropriate manufacturing processes used in the production of the trolleys **and** explain why each one is suitable. **6**

(c) Describe how function has influenced the design of the trolleys. **5**

(d) Explain the benefits **and** drawbacks of using standard components in the assembly of the trolleys. **4**

(e) Explain the benefits of using physical modelling in the design of products such as the trolleys. **4**

[Turn over

MARKS

SECTION 2 — 55 marks

Attempt ALL questions

2. A child's safety helmet is shown below.



A range of thermoplastics have been used in the manufacture of the safety helmet.

(a) Identify **one** thermoplastic that may have been used in the manufacture of the safety helmet **and** explain why it is suitable. **3**

The needs of the target market must be considered when designing products such as the child's safety helmet.

(b) Explain how the needs of the target market may have influenced the design of products such as the child's safety helmet. **4**

The manufacturer must ensure the safety helmet is fit for purpose.

(c) Describe an evaluation technique that could be used to ensure the safety helmet is fit for purpose. **2**

MARKS

3. Clients can provide designers with an open brief.

(a) Describe **one** advantage and **one** disadvantage of an open brief. **2**

A product design specification is produced after analysis of the brief.

(b) Describe the purpose of the product design specification **and** outline the types of information it may contain. **3**

Designers may use idea generation techniques to produce initial ideas.

(c) Outline the key stages of an idea generation technique and explain why it would be used to help produce initial ideas. **3**

Designers use a variety of graphic techniques during the design process.

(d) Describe how graphic techniques are used during the design process. **4**

(*You should refer to a variety of graphic techniques in your answer.*)

4. The successful launch of a product requires effective marketing techniques to be used.

(a) Describe marketing techniques that could be used to ensure the successful launch of a product. **4**

Towards the end of a product's life cycle, sales will start to decline.

(b) Describe the steps a company may take to delay the decline in product sales. **2**

[Turn over

MARKS

5. An air fryer is shown below.



(a) Describe how ergonomics has influenced the design of the air fryer. **6**

(*You should cover all **three** areas of ergonomics in your answer.*)

Products such as air fryers were developed in response to market pull, to ensure consumer demand was met.

(b) Describe how the design of products has been influenced by market pull. **2**

(*You should refer to products you are familiar with in your answer.*)

Trademarks and patents are examples of Intellectual Property Rights (IPR).

(c) Outline the advantages of using Intellectual Property Rights (IPR) to protect products. **2**

MARKS

6. A variety of methods can be used to identify materials such as woods, metals and plastics.

(a) Describe methods that could be used to help identify different types of materials. **4**

Manufacturing processes have a number of identifying features.

(b) Outline **two** identifying features of blow moulding. **2**

Production and planning systems are used in the development and manufacture of products.

(c) Describe how the following production and planning systems can be used in the development and manufacture of products:

- Gantt charts
- Just-in-Time (JIT) production. **4**

(*You should refer to both Gantt charts and Just-in-Time (JIT) production to attract full marks*.)

7. Design teams have a responsibility to consider the environmental impact of their products.

Describe how consideration for the environment has influenced the design, manufacture, use and disposal of products. **8**

[END OF QUESTION PAPER]

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